

Xinyi Wang

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Education

- **Postdoctoral Fellow**, Robotics Department, University of Michigan 09/2024-Now
Supervisor: [Prof. Dimitra Panagou](#)
- **Postdoctoral Fellow**, Hong Kong Centre for Logistics Robotics 10/2023-09/2024
- **Ph.D. Degree**, Department of Mechanical & Automation Engineering, The Chinese University of Hong Kong 08/2019-09/2023
Supervisor & Co-supervisor:
[Prof. Ben M. Chen](#) (Academy of Engineering Fellow, Singapore)
& [Prof. Jie Huang](#) (Foreign Member of the Chinese Academy of Engineering (CAE))
Medium of Instruction: English
- **B.Eng. Degree**, School of Aerospace Engineering, Xiamen University 09/2015-06/2019

Research Interests

Optimization and Control Theory, Multi-Agent Systems, Safe Reinforcement Learning, Motion Planning.

Selected Research Projects

Socially Aware Safe Navigation for Heterogeneous Multi-Agent Systems.

Toyota Research North America. Total: 150,000 USD. Core Member. 09/2024-Now

- Developed a risk-adaptive safety controller based on CVaR Control Barrier Functions in uncertain, human-crowded environments; integrated the controller as a safety filter for RL policies.
- Developed a differentiable risk-aware optimization layer that models multi-modal human behaviors and quantifies interaction risk via CVaR, seamlessly integrated into the end-to-end RL training pipeline for adaptive risk sensitivity.
- Validated on real-world pedestrian trajectory datasets, demonstrating robust generalization and safety under out-of-distribution scenarios where standard RL methods fail.

Enabling Autonomous, Persistent, and Adaptive Mobile Observational Networks Through Energy-Aware Dynamic Coverage.

National Science Foundation (NSF). Total: 1,000,000 USD. Core Member. 04/2025-Now

- Developed theoretical guarantees for the spatiotemporal estimation problem in multi-agent sensing networks.
- Delivered a practical pre-deployment design tool for sensor network sizing and resource allocation.

Aerial-Marine Collaborative Heterogeneous Multi-Agent Systems with Digital Twin Technology for Smart Oceans.

General Research Fund, Hong Kong. Total: 1,134,931HKD. Co-I & Write proposal. 07/2023–08/2024

- Co-investigator and proposal co-author, initiating the project based on my research on fully decentralized, deadlock-free, and collision-free multi-agent search and coverage.
- Validated the framework using only local sensing and 3D point-cloud data, enabling scalable operation in forest-like, obstacle-dense outdoor environments.

Development of an Integrated Autonomous Building Inspection and Information Management.

InnoHK Center for Logistics Robotics, Hong Kong. Core Member. 07/2022-07/2023

- Developed a fast, safe collaborative task-assignment framework for multi-agent unmanned systems in infrastructure inspection, forming the basis of a successful 2-year GRF-funded proposal (HKD 1,369,862).
- Applied the framework to a real-world industrial infrastructure inspection using a three-UAV system in Shatin, Hong Kong, achieving significant reductions in inspection time and operational risk.

Collaborative Search and Pursuit-evasion for Unmanned Systems in Cluttered Environments.

General Research Fund, Hong Kong. Total: 1,382,623 HKD. Write proposal & Core Member. 11/2021-07/2022

- Developed an efficient cooperative encirclement strategy for pursuing an adversarial evader in unbounded, cluttered environments.
- Validated the strategy in real-time real-world experiments with four mobile robots, including scenarios with autonomous and human-controlled evaders.

Advanced Motion Planning Techniques for the Cooperation of Multi-agent Systems.

General Research Fund, Hong Kong. Total: 815,601 HKD. Write proposal & Core Member. 09/2019-11/2022

- Designed a sampling-based motion planning algorithm with real-time adaptation and deployed it on real-time multicopter platforms using onboard computation.
- Designed a neural-accelerated, distributed MPC trajectory planner for multi-quadrotor systems, and deployed it on the Crazyflie platform.

Selected Publications

(# = equal contribution, * = corresponding author)

● Preprints

- [P1] **X. Wang**#, D. R. Agrawal#, and D. Panagou, Kalman-Bucy Filtering with Randomized Sensing: Fundamental Limits and Sensor Network Design for Field Estimation, *Automatica*, 2025 (Under Review).
- [P2] J. Chen, Y. Luo, **X. Wang**, J. Li, M. Guo. AMBUSH: Collaborative Capture in Complex Environments with Neural Acceleration. *IEEE Transactions on Automation Science and Engineering (T-ASE)*, 2025 (Under Review).
- [P3] **X. Wang**, T. Kim, B. Hoxha, G. Fainekos, D. Panagou, Learning to Adapt Risk via Differentiable CVaR Barrier Functions. *IEEE Robotics and Automation Letters (RA-L)*, 2026 (Under Review).

● Journal papers

- [J1] V. S. Chipade, **X. Wang**, D. Panagou. IDCAIS: Inter-Defender Collision-Aware Interception Strategy against Multiple Attackers. *IEEE Transactions on Control of Network Systems (TCNS)*, 2025.
- [J2] **X. Wang**, L. Xi, Y. Ding, B. M. Chen. Distributed Encirclement and Capture of Multiple Pursuers with Collision Avoidance. *IEEE Transactions on Industrial Electronics (TIE)*, 2023.
- [J3] C. Gao, **X. Wang***, X. Chen, B. M. Chen. A Hierarchical Multi-UAV Cooperative Framework for Infrastructure Inspection and Reconstruction. *Control Theory and Technology (CTT)*, 2023.
- [J4] **X. Wang**, L. Xi, Y. Chen, S. Lai, F. Lin, B. M. Chen. Decentralized MPC-based Trajectory Generation for Multiple Quadrotors in Cluttered Environments. *Guidance, Navigation and Control (GNC)*, 2021.
- [J5] C. Gao, **X. Wang**, R. Wang, Z. Zhao, Y. Zhai, X. Chen, B. M. Chen. A UAV-based Explore-then-exploit System for Autonomous Indoor Facility Inspection and Scene Reconstruction. *Automation in Construction (AIC)*, 2022.
- [J6] L. Xi, **X. Wang**, L. Jiao, S. Lai, Z. Peng, B. M. Chen. GTO-MPC-Based Target Chasing Using a Quadrotor in Cluttered Environments. *IEEE Transactions on Industrial Electronics (TIE)*, 2021.

● Conference papers

- [C1] **X. Wang**, T. Kim, B. Hoxha, G. Fainekos, D. Panagou, Safe Navigation in Uncertain Crowded Environments

Using Risk Adaptive CVaR Barrier Functions. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025.

[C2] C. Wang, **X. Wang**, Y. Dong, L. Song, X. Guan. Multi-Constraint Safe Reinforcement Learning via Closed-form Solution for Log-Sum-Exp Approximation of Control Barrier Functions. Learning for Dynamics & Control Conference (L4DC), 2025.

[C3] **X. Wang**, J. Xu, C. Gao, Y. Chen, J. Zhang, C. Wang and B. M. Chen, Sensor-based Multi-Robot Search and Coverage with Spatial Separation in Unstructured Environments. IEEE International Conference on Robotics and Automation (ICRA), 2024.

[C4] **X. Wang**, Y. Ding, Y. Chen, R. Han, L. Xi, B. M. Chen. OA-ECBVC: A Cooperative Collision-free Encirclement and Capture Approach in Cluttered Environments. IEEE Conference on Decision and Control (CDC), 2023.

[C5] Y. Chen, R. Wang, **X. Wang***, B. M. Chen. Sampling-based Path Planning under Temporal Logic Constraints with Real-time Adaptation. IEEE International Conference on Robotics and Automation (ICRA), 2023.

[C6] **X. Wang***, C. Gao, X. Chen, B. M. Chen. Fast and Secure Distributed Multi-Agent Coverage Control with an Application to Infrastructure Inspection and Reconstruction. Chinese Control Conference (CCC), 2023, Best Paper Award Finalist.

[C7] C. Gao, **X. Wang***, X. Chen, B. M. Chen. An Active Search Strategy with Multiple Unmanned Aerial Systems for Multiple Targets. International Conference on Unmanned Aircraft Systems (ICUAS), 2024.

[C8] L. Xi#, **X. Wang#**, Y. Ding, Y. Wei, Z. Peng, B. M. Chen. Quadrotor Trajectory Planning for Visibility-aware Target Following. IEEE International Conference on Robotics and Biomimetics (ROBIO), 2021.

[C9] Y. Chen, **X. Wang**, R. Wang, Z. Guo, S. Lai, B. M. Chen. An Interactive System for Multiple-task Linear Temporal Logic Path Planning. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023.

[C10] R. Wang, Z. Guo, Y. Chen, **X. Wang** and B. M. Chen, Air bumper: A collision detection and reaction framework for autonomous MAV navigation. IEEE International Conference on Robotics and Automation (ICRA), 2024.

• Patents

[1] B. M. Chen, Y. Chen, R. Wang, **X. Wang**, Xi Chen. A Human-in-the-Loop System for Mobile Robot Task Planning. Patent of the United States, No. 63/450,314, Mar. 6, 2023.

Technical Skills

- Experienced in Python, PyTorch, C++, MATLAB, Julia, Robot Operating System (ROS), Gazebo, Git

Courses & Teaching Experience & Mentorship Experience

Courses:

- IERG 5350 Reinforcement Learning, instructed by Professor Bolei Zhou.
- ENGG 5501 Foundations of Optimization, instructed by Professor Anthony Man-Cho So.
- ELEG 5491 Introduction to Deep Learning, instructed by Professor Hongsheng Li.

Teaching:

- ROB 416 Multi-Robot Systems, **Guest Lecturer**, instructed by Professor Dimitra Panagou.
- EECS 560 Linear Systems Theory, **Guest Lecturer**, instructed by Professor Necmiye Ozay.
- MAEG 3080 Fundamentals of Machine Intelligence, **Teaching Assistant**, instructed by Professor Juan Luis.
- ENGG 5402 - Advanced Robotics, **Teaching Assistant**, instructed by Professor Fei Chen.

Training:

- [U Michigan, STEM Postdoc Teaching Course](#) (Ongoing) 2026
- Professional Development Course for Teaching Assistant, CUHK 2021

Mentorship:

- Summer Undergraduate Research in Engineering (Ongoing) 2026
- Guided new Ph.D. students in getting started with research and technical paper writing 2022-2024
- Guided Master students on MAEG 5910 M.Sc. Project 2023

Awards & Honors

UMPDA Postdoc Conference Award	10/2025
Postdoctoral Scholarship, the Innovation and Technology Fund of Hong Kong	10/2023-09/2024
Professor Charles K. Ko Student Creativity Award, CUHK	06/2023
Research Postgraduate Studentship, CUHK	2019-2023
Outstanding Graduates, XMU (Top 2%)	06/2019
Merit Student, XMU (Twice, Top 8%)	12/2016 & 12/2017
Third Prize in RoboMaster 2017 (The 16th National College Students Robotics Contest)	08/2017
Xiamen International Bank Scholarship, XMU (Top 4%)	04/2017

Academic Services**• Reviewer of Journals**

- IEEE Transactions on Robotics (TRO)
- IEEE Transactions on Automatic Control (TAC)
- IEEE Transactions on Cybernetics
- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Intelligent Transportation Systems (T-ITS)
- International Journal of Robust and Nonlinear Control

• Reviewer of Conferences

- Associate Editor of IEEE International Conference on Control and Automation (ICCA)
- Robotics: Science and Systems (RSS)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- IEEE Conference on Decision and Control (CDC)
- American Control Conference (ACC)
- IEEE International Conference on Unmanned Aircraft Systems (ICUAS)